The Lunar Polesitter

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Here-to-fore, sailcraft mission and system studies have focused on sailcraft applications in support of NASA's science missions and, in a few studies, on the needs of other federal agencies such as the National Oceanic and Atmospheric Administration (NOAA) and Department of Defense (DoD). These studies have identified numerous promising applications for solar sails, leading NASA to support proposal efforts for three NASA New Millennium Program (NMP) flight demonstration opportunities (the Space Technology-5, -7, and -9 opportunities) as well as an extensive three-year ground development program in FY 2003–2005 sponsored by the NASA In-Space Propulsion Technology (ISPT) Program. What has not been done to date, however, is to investigate how the technology might also benefit the nation's (and NASA's) emerging interest in the Human Exploration Initiative (HEI). This paper reports on the first effort to address this shortfall in mission applications studies in support of HEI: the use of solar-sail-propelled Lunar Polesitter spacecraft which make use of the natural properties of the Earth-Moon L2 point and solar sail propulsion to enable their positioning near the Lunar poles to serve as communications relay stations. Suitably positioned, such spacecraft enable continuous communications to and from the Earth from any point on the lunar far side. The paper shows that a viable sailcraft system design exists permitting station-keeping of a Lunar Polesitter relay station at 40 Lunar radii from the Moon in the anti-Earth direction, displaced 6–8 Lunar radii below the Earth-Moon plane.

Nomenclature

a_0 = characteristic acceleration

I. Introduction

HERE-to-fore, sailcraft mission and system studies have focused on sailcraft applications in support of NASA's science missions and, in a few studies, on the needs of other federal agencies such as the National Oceanic and Atmospheric Administration (NOAA) and Department of Defense (DoD). These studies have identified numerous promising applications for solar sails,[1] leading NASA to support proposal efforts for three NASA New Millennium Program (NMP) flight demonstration opportunities (the Space Technology-5 [Ref. 2], -7, and -9 opportunities) as well as an extensive three-year ground development program in FY 2003-2005 sponsored by the NASA In-Space Propulsion Technology (ISPT) Program. What has not been done to date, however, is to investigate how the technology might also benefit the nation's (and NASA's) emerging interest in the Human Exploration Initiative (HEI). This paper reports on the first effort to address this shortfall in mission applications studies in support of HEI: the use of solar-sail-propelled Lunar Polesitter spacecraft which make use of the natural properties of the Earth-Moon L2 point and solar sail propulsion to enable their positioning near the lunar poles to serve as communications relay stations. So positioned, a Lunar Polesitter spacecraft becomes the equivalent of an Earth geosynchronous

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communications satellite only fixed in space in a more-nearly polar region of the Moon rather than over the Moon's equator, enabling continuous pointing of lunar-surface-asset communications antennas to a fixed site in space for data relay to the Earth.

Important to note is the name "Lunar Polesitter" for the relay satellite design that emerged from the research documented in this paper is a misnomer. This is because, although the study team's goal upon embarking on the study was to develop the design of a "true" Polesitter spacecraft, that is, of a lunar relay satellite with sail capabilities that would permit it to hover precisely over a lunar pole to serve its communications functions, the findings of the research showed the goal was not achievable and dictated the need to look in another direction. Specifically, this was the result of the finding that balancing the acceleration demands (principally due to the Moon itself) on a spacecraft so positioned geometrically and at the same time close enough to the Moon for the communications system to support data rates potentially useful to the HEI community was not viable with either current sail propulsion technology or any reasonable projection of that capability in the timeframe of interest to the HEI community. So the team was led in another direction, specifically in the direction of exploring other potentially useful locations near the Moon including those that might exploit the properties of the Earth-Moon L1 and L2 points, or, more properly, the Earth-Moon L1 and L2 regions. This led to the ultimate finding reported herein, that a location on the lunar far side only slightly displaced southward from the Earth-Moon plane and nearest the Earth-Moon L2 point could both satisfy the mission's geometry requirements and be achieved with reasonable projections in expected solar sail performance.

II. The Lunar and Mars Communications Challenges

Figure 1 summarizes the Lunar and Mars communications challenges facing the Human Exploration Program, or, as it is currently better known, the Constellation Program, that were utilized by the author's JPL Team in designing the Lunar Polesitter relay satellite and mission architecture. The Team derived the challenges from conversations with NASA at the time the work described herein was being performed, during the period October 2006 through December 2007, focusing on the identification of "core" Program requirements most likely to survive the evolving requirements generation process and the longer test of time. As such, although the five challenges neither then nor now, in a formal sense, represent Constellation Program requirements, they do constitute the kernal of requirements the Team believes will continue to guide the Program's communications needs.

III. The Polesitter Advantages

Figure 2 summarizes the seven key advantages offered by the Lunar Polesitter compared to relay satellites and relay satellite architectures using conventional orbits. While all the advantages are important, the author believes the advantages listed second, third, and fifth are the most compelling. These advantages are: continuous coverage;

- **Provide basic navigation and communication mission support for the initial US/NASA-developed launch vehicle and lunar lander capabilities developed for the human exploration of the Moon:**
  - Launch vehicle architecture: Crew Exploration Vehicle (CEV). Not to include the Ares launch vehicles and Earth Departure Stage.
  - Lunar lander: Ascent vehicle, descent vehicle, basic habitation at or near the lunar south pole.
- **Provide for easy augmentation of communications services at the Moon.**
  - Higher bandwidth.
  - Extension of services beyond the initial south pole station to include human and robotic exploration 100 km to 200 km beyond the initial site and ultimate extension to complete coverage of the lunar far side in support of desired science community objectives.
- **Provide for portability of the communications architecture to Mars.**
- **Provide for potential international cooperation in meeting the communication and navigation objectives.**
- **Satisfy the following key requirements in support of the initial capabilities:**
  - 10 Gb/day from the lunar surface.
  - 8 kbps relay-to-CEV; 28 kbps CEV-to-relay.

Figure 1. The Lunar and Mars communications challenges.

lower resource demand on surface assets; and lower cost. The paragraphs below provide selected additional observations and details pertaining to the seven advantages.

A. Easily Satisfies the Key Requirements in Support of the Initial Capabilities

As noted in Figure 2, the Polesitter mission-system design, which includes all elements of the architecture, including the relay satellite system design, sail propulsion system design, communications subsystem design, and mission design, easily satisfies NASA's communications system requirements for the initial capabilities. This includes the requirement for a data volume of 10 Gb per day from the Moon with the system design capable of supporting up to 13 Gb per day and the requirement to support 8 kbps relay-to-CEV and 28 kbps CEV-to-relay links. Key to the system's ability to meet the data volume per day requirement and remain competitive with the alternatives whose conventional orbits bring them closer to the lunar surface is the Polesitter's ability to trade communications contact time for link rate. This means the Polesitter with its continuous visibility to the Earth but greater distance from the Moon can utilize contact time to remain competitive with the alternatives on a data volume per day basis without the need for an outsized communications subsystem and antenna to match the alternatives' higher link-rate capability. It also keeps the cost of the Polesitter communications subsystem competitive with the alternatives.

B. Continuous Coverage

Unique to the Polesitter concept is the capability to provide continuous communications to and from the Earth and to do so with only one satellite and without the need to re-point communications antennas on the surface of the Moon. This enables an always-available safety net for astronaut crews to mission controllers and for real-time ground support for astronaut operations on the lunar surface. It also enables lower resource demand on surface assets, greater operational simplicity, and lower cost, as discussed below.

C. Lower Resource Demand on Surface Assets

Lower resource demand on surface assets is a direct consequence of the fact surface assets point at a single point in space and do not require acquisition and tracking systems, mechanisms, and software to acquire and track the relay satellite as do the alternatives. The analogy is precisely the same as Direct TV, with the same advantages. This

- **Easily Satisfies the Key Requirements in Support of the Initial Capabilities:**
  - 13 Gb/day capability from the lunar surface vs 10 Gb/day requirement.
  - 8 kbps relay-to-CEV and 28 kbps CEV-to-relay capabilities, matching the requirements.
- **Continuous Coverage:** Relay satellite is in continuous view of lunar surface assets and the Earth, enabling continuous coverage from the Earth. Alternatives have dark periods.
- **Lower Resource Demand on Surface Assets:** No requirement for surface asset acquisition and tracking systems and mechanisms. Simpler, lower-cost, lower-mass, surface assets.
- **Greater Operational Simplicity:** Surface assets point continuously at a fixed point in space. Alternatives require surface assets to continuously acquire, track, and re-acquire relay satellites.
- **Lifetime Not Propellant Limited:** Polesitter requires no propellant. Alternatives require propellant.
- **Lower Cost:** Polesitter relay concept requires only one satellite for continuous coverage of the lunar south pole, two for north and south pole coverage. Alternatives require up to six satellites for continuous coverage. Low launch mass of 235-255 kg may permit piggyback launch, launch on a smaller, lower-cost launch vehicle than the alternatives.
- **More Attractive for International Participation**

Figure 2. The Polesitter advantages.

means simpler, lower mass, lower cost surface assets. Perhaps very importantly, it eliminates the need for robust, potentially expensive tracking mechanisms that can operate reliably for extended periods in the lunar dust environment.

D. Greater Operational Simplicity

Greater operational simplicity is another immediate advantage of the fact surface assets can point at a single point in space. This means the elimination of complex tracking software and the associated software maintenance, updates, and uploads.

E. Lifetime Not Propellant Limited

Utilizing solar photon pressure for station keeping, the Polesitter's operational lifetime, unlike the alternatives, is largely independent of propellant. This frees mass for other functions, as well as eliminating the need for propellant mass management.

F. Lower Cost

The Polesitter's cost advantage compared to the alternatives is derived principally from two aspects of the Polesitter mission-system architecture: first and foremost, and compellingly, the fact only one satellite is required to meet the communications requirements; and second, the fact the architecture permits surface assets to operate without acquisition, tracking, and pointing mechanisms. The first advantage affects overall program infrastructure costs; the second affects the costs of potentially all (if not all) users of the infrastructure. As noted in Figure 2, the low mass of the satellite also permits potential launch vehicle cost savings.

G. More Attractive for International Cooperation

As discussed in detail in paragraph **V**, world interest in solar sail technology is already high, with the Polesitter's use of renewable sources of energy offering a natural advantage upon which to build world public interest and international participation.

IV. The Polesitter Concept

The two paragraphs that follow provide an overview description of the Polesitter mission and relay satellite design. In addition, details pertaining to the technology readiness and heritage of the critical elements of the relay satellite are provided.

A. Mission Design Description

The Polesitter mission design can be thought of in two phases. The first phase involves launch, getting to the Moon, and getting into lunar orbit, and the second phase involves station keeping at a particular desired operational station location near the Moon.

The first phase uses highly-conventional, well-known, strategies for launch, Earth-Moon transfer, and lunar orbit insertion, with launch of the Polesitter satellite on a small-to-medium-size launch vehicle, ballistic transfer to the Moon, and lunar orbit insertion utilizing conventional chemical propulsion. The phase ends with deployment of the solar sail and jettison of the carrier vehicle which, as is discussed in more detail in paragraph **IV-B**, houses the motor for the orbit insertion burn. Conventional as it is, phase one will not be discussed further in this paper.

The second phase, in contrast, utilizes an unconventional, previously unsubstantiated, approach using solar sail propulsion and the properties of the Earth-Moon L2 point to station keep the satellite at a point near the Moon useful for it to function as a relay for data acquired at or near the lunar poles or from the lunar far side to the Earth. Where the challenge lay for the team was in finding whether there was an achievable location for the satellite, that is, a location achievable given the maturity (and acceleration performance capability or characteristic acceleration, a) of contemporary solar sail technology (or at least technology that could be achieved in the timeframe of interest), that was also useful from the standpoint of mission geometry (that is, the position of the relay with respect to the Moon and Earth and its distance from the Moon) and achievable rely communications subsystem performance with regard to data rate and/or data volume returned to the Earth. In short, although it seemed theoretically possible a solar sail could provide the acceleration and the attitude control authority necessary to modulate that acceleration to balance the Moon's, Earth's, and Sun's gravitational forces to achieve and maintain a stationary location somewhere near the Moon, it remained for the team to design the relay satellite, design and size the sail, design the communications subsystem and calculate its performance, determine the masses of the system elements, and create the mission design tools and techniques to determine where such a location might be, how a satellite placed there would behave

under the gravitational forces, and what the characteristics of the satellite would have to be to make the location useful. In the course of the investigation, the team performed analyses and trades that included, but were not limited to, characterization of the gravity wells around the Moon and, in particular, near both the Earth-Moon L1 and L2 points, investigation of the effect of the Earth's and Sun's gravitational influence on a satellite in those regions over an Earth year, and investigation of potential relay satellite distances from the Moon from a few up to 80 lunar radii (RL) with regard to required sail and relay satellite communications subsystem performance. What the team discovered is: 1) there is a useful location and it is achievable and maintainable with modest improvement in contemporary solar sail technology, 2) the mission geometry works, and 3) the communications subsystem performance is adequate to meet the NASA customer's initial needs for communications performance. Figure 3 illustrates the mission geometry that was ultimately defined, and the paragraph below discusses some of the details of that geometry and its consequences for the design of the Polesitter mission. Left to discuss in other papers which are currently being written by the author's team are other complementary findings and details of the mission analysis tools and techniques.

As noted in the preceding paragraph, Figure 3 illustrates the Polesitter mission geometry. The best solution found for the initial position of the Polesitter at the Moon was on the lunar far side in the vicinity of the Earth-Moon L2 point slightly displaced southward from the Earth-Moon plane, that is, at location: X = –44 lunar radii (RL), Y = 0 RL, Z = –7 RL. The figure shows an example representative of that solution with initial sailcraft position at location: X = –37 RL, Y = 0 RL, Z = –7.25 RL This places the relay satellite in continuous view of manned and robotic assets at or near the lunar south pole and in continuous view of the Earth for the relay-to-Earth link. Stunningly and fortuitously, what the results also showed was that the sailcraft required no active maneuvering to maintain a usefully-fixed station location for its use as a relay. What is meant by a "usefully-fixed" location is the sailcraft was found to change its location by a small enough amount due to the drift caused by the combined effects of the Moon's, Earth's, and Sun's gravity that it can continuously, even without active sailcraft control of its location, remain within the field-of-view of a lunar surface asset equipped with a simple fixed-azimuth and fixed-elevation shaped-beam antenna. Figure 4 illustrates this, showing the path of the relay satellite in the field-of-view of a surface asset's notional 60 deg × 15 deg shaped-beam antenna. The results further showed that the least drift was obtained for fixed cone angles (defined as the angle between the sailcraft-Sun line and sail plane normal) in the range of 55 deg to 60 deg (with 55 deg preferred due to sail design constraints) and a sailcraft characteristic acceleration (a_0) of 1.055 mm/s2 (compared to a sailcraft capability of 1.2 mm/S2 for the sailcraft design

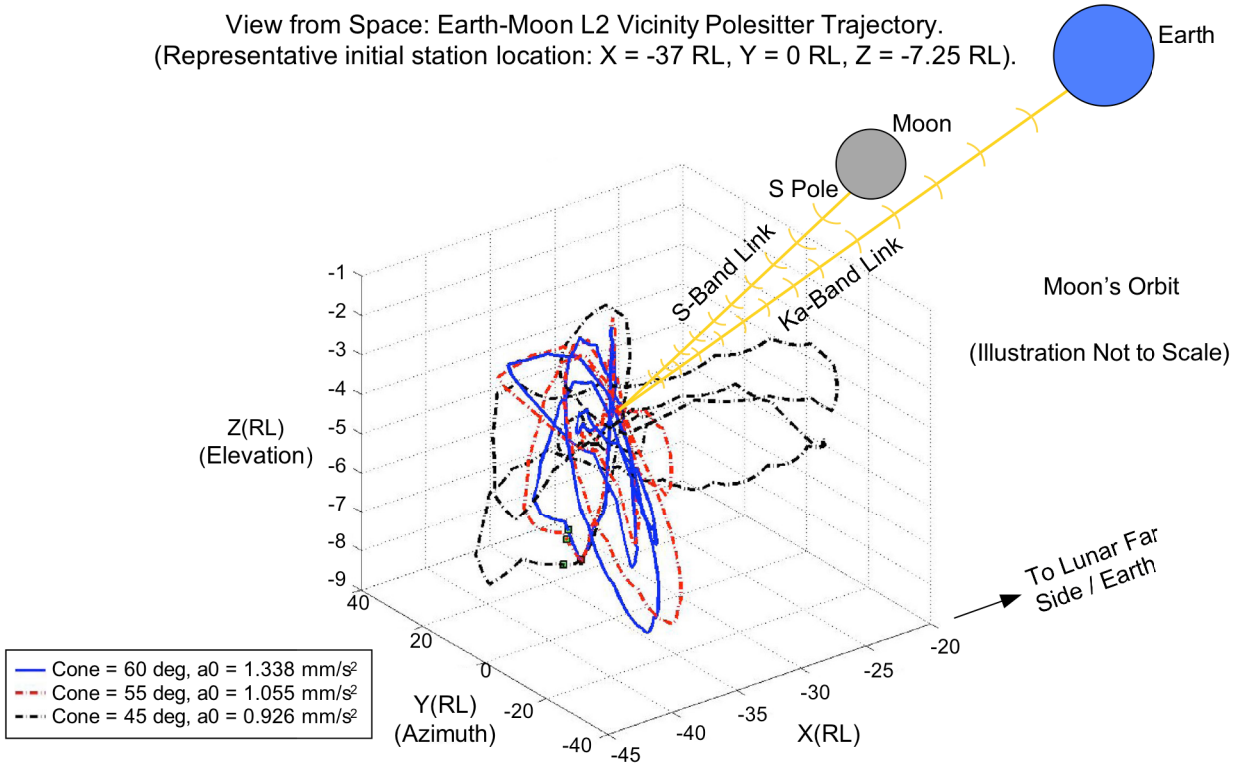


Figure 3. Polesitter mission geometry viewed from space.

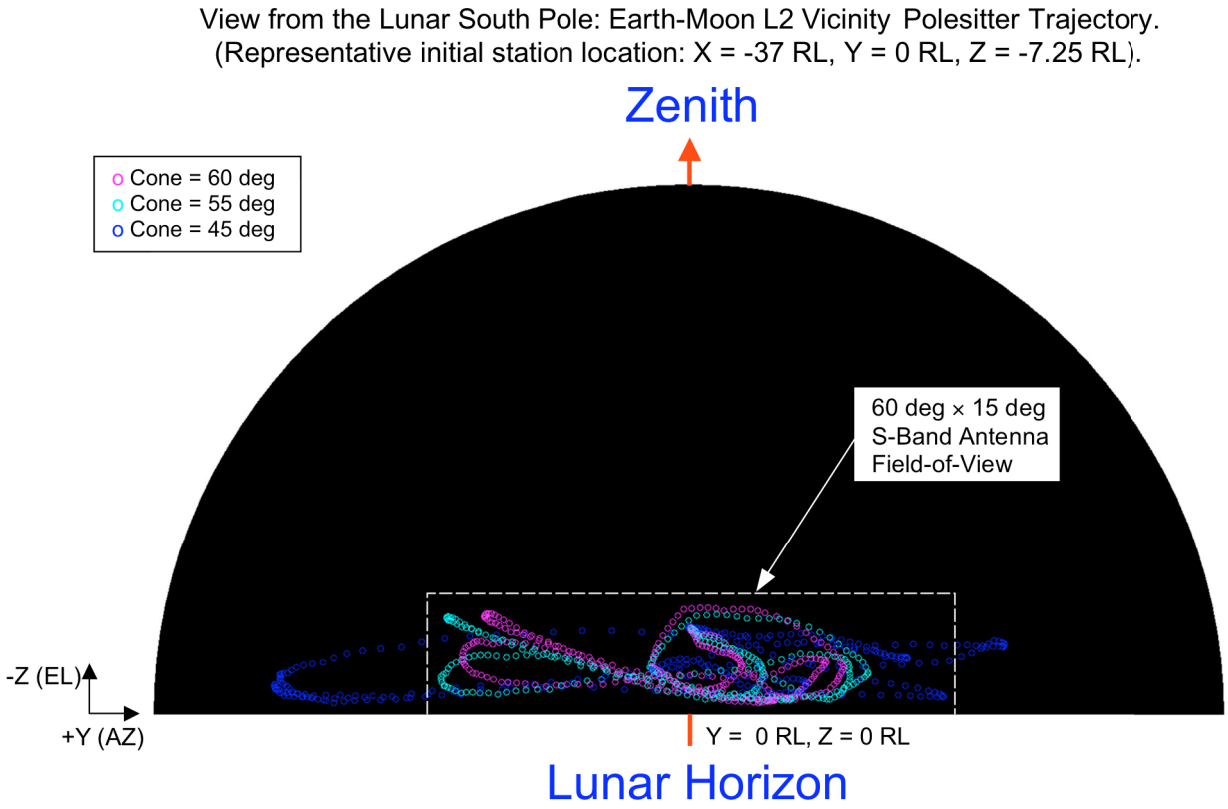


Figure 4. Polesitter trajectory viewed from the lunar south pole.

documented herein). As also noted in paragraph **IV-B**-*1*, the mission geometry dictates the need for continuous gimbaled antenna tracking of the Moon and Earth to maintain, respectively, the S-band lunar-surface-to-relay and Ka-band relay-to-Earth communications links, with maintenance of the links also requiring twice per lunar cycle 180-degree repositioning of a relay-bus-mounted "yoke" on which the antennas are mounted. It is observed a second Polesitter located near the lunar north pole could enable complete lunar far side coverage, continuous Earth-to-lunar-polar-orbiting CEV coverage, and/or redundancy for a south pole Polesitter, were it to be maneuvered, using its control vanes, to the south pole region. Nominal lifetime for the satellite is five years.

B. Relay Satellite Description

The relay satellite (as communications engineers see it) or sailcraft (as propulsion engineers see it) has a mass of 195 kg with 20% contingency. It is comprised of two elements: the relay bus and the Sail Propulsion System (SPS). The relay bus houses the standard elements required by a satellite or spacecraft for basic functionality and the SPS provides the propulsion for station keeping and attitude control. Three figures serve to illustrate the concept. Figure 5 shows the relay satellite with the relay bus at the center of the deployed solar sail as it would appear in operational configuration at the desired on-station location. Figures 6a and 6b, respectively, show the relay satellite with its communication antenna "farm" stowed and antenna farm deployed, both for the L1 station option. And Figure 7 shows the relay satellite (without antennas) as it would appear immediately following separation from its "carrier" vehicle. The carrier vehicle is not itself part of the relay satellite but is attached to the relay satellite at launch and provides three critical support functions for the relay satellite: 1) accommodation of launch loads, 2) propulsion for relay satellite insertion into the desired operational station at the Moon, and 3) housing for the sail deployment hardware. Once the relay satellite is inserted into the operational station at the Moon and the sail is deployed, the carrier vehicle is jettisoned to reduce relay satellite mass and improve sailcraft performance to enable maintenance at the desired location with a technologically achievable sail size. The carrier vehicle has a mass of 40 kg to 60 kg, with 20% contingency, including propellant for mid-course delta-V and orbit insertion. By way of definition, the carrier vehicle and the relay satellite comprise the flight vehicle (or Polesitter flight vehicle), that is, the vehicle configuration at launch that makes the transit to the Moon. The Polesitter flight vehicle mass (that is, the vehicle launch mass) is 235 kg to 255 kg with 20% contingency.

1. Relay Bus

The relay bus has a mass of 135 kg with 20% contingency including 64 kg for the critical communications subsystem. As noted in the paragraph above, the relay bus contains the standard elements required by a satellite or spacecraft for basic functionality with nothing unusual - aside from the communications subsystem - required to support the Polesitter mission. As such, it will not be dwelled on herein. In contrast, while introducing no new technology, the communications subsystem requires a novel design to accommodate the Polesitter mission, as discussed below.

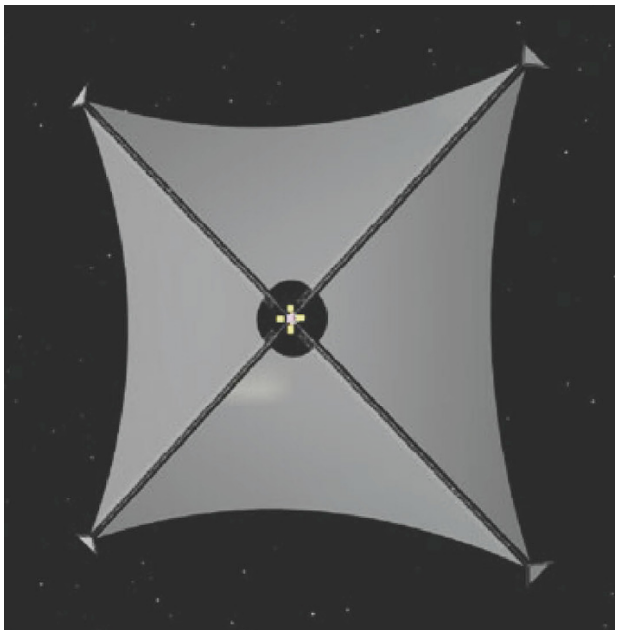


Figure 5. Polesitter relay satellite in operational configuration. Relay bus is at the center of the deployed sail. Sail attitude control vanes are at the sail tips.

As noted above, the Polesitter communications subsystem is uniquely designed to support the Polesitter mission. The subsystem consists of two antennas. The first is a 3.4-m-diameter two-axis-gimbaled S-band deployable mesh antenna (notionally the Northrop Grumman AstroMesh antenna, as shown in Figure 8) for the relay to/from lunar surface and relay to/from CEV links. The second a 0.2-m-diameter two-axis gimbaled Ka-band fixed dish antenna for the relay to/from Earth trunk link. The S-band lunar-surface-to-relay satellite link operates at a nominal 150 K bps uplink data rate with a minimum link margin of 3 db (as have all the links described herein) with selectable lower data rates and supports a data volume of 13 Gb per day from the lunar south pole to the Earth adequate to meet NASA's notional requirement of 10 Gb per day from the lunar surface. The relay-to-CEV S-band link operates at 8 K bps and

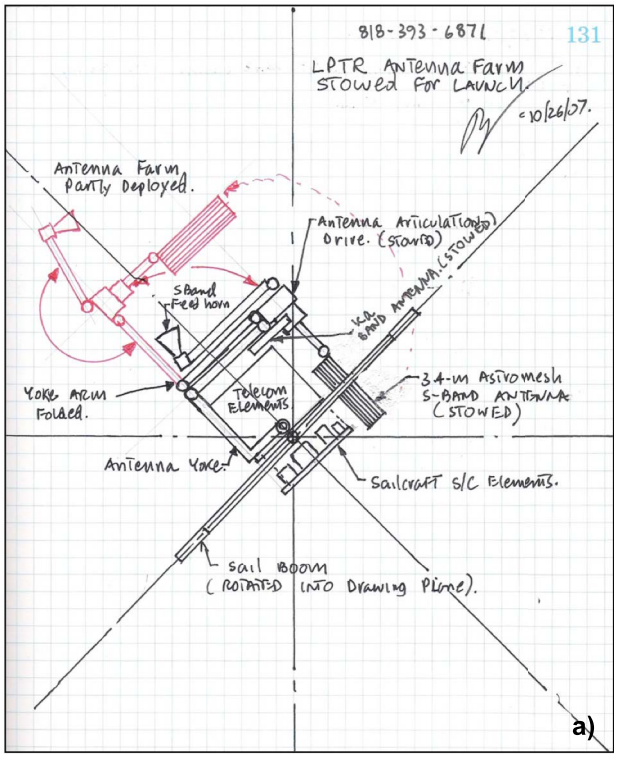


Figure 6a. Relay satellite with antenna "farm" stowed (L1 station option).

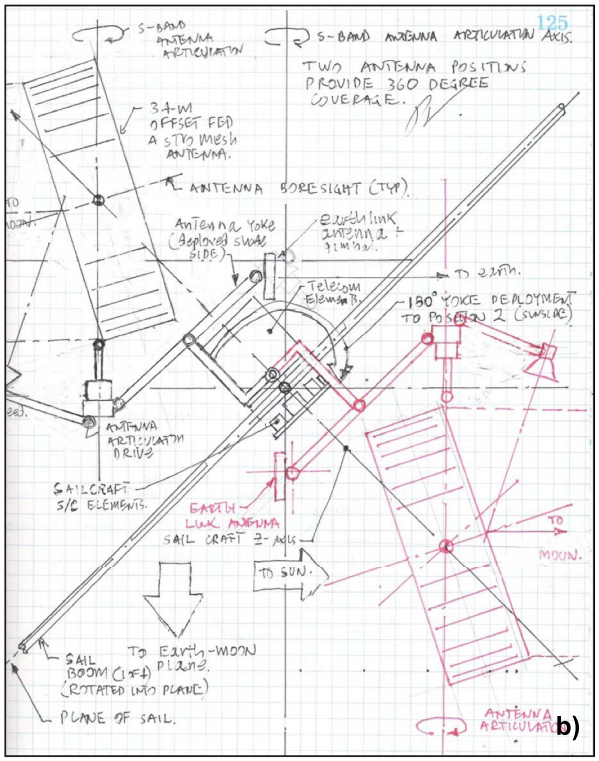


Figure 6b. Relay satellite with antenna "farm" deployed (L1 station option).

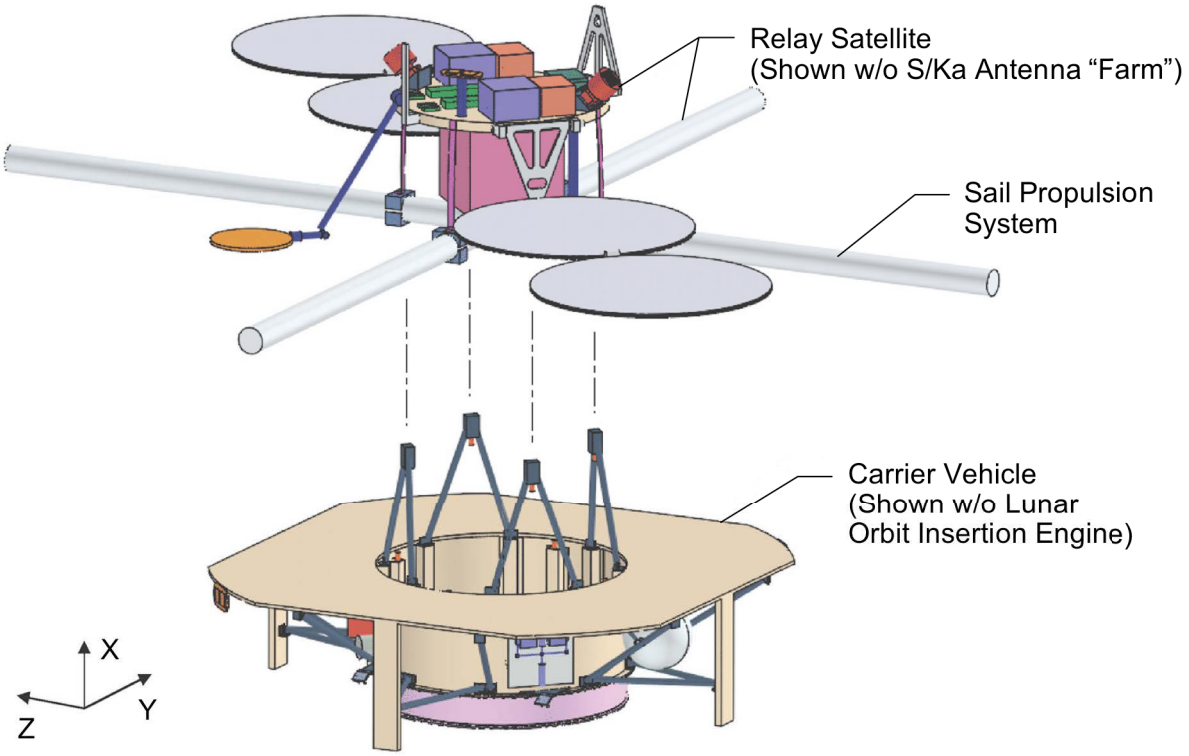


Figure 7. Relay satellite with carrier vehicle.

CEV-to-relay link at 28 K bps, assuming the CEV S-band omni antenna and link parameters documented in the Constellation Master Link Book. The S-band lunar-surface-to-relay satellite link design accommodates data from two lunar surface assets simultaneously (and could be designed to accommodate more, if desired). The Ka-band trunk link operates at a nominal downlink data rate to Earth of 150 K bps in bent-pipe mode and 1 M bps in store and forward mode. Critical to maintaining the lunar-surface-to-relay satellite link performance and realization of the communications architecture described herein, in which the surface assets need not re-point their antennas to track the relay satellite, is the requirement for the surface assets to have notional 60 deg × 15 deg field of view shaped-beam antennas with presumed antenna area equivalent to that of a 0.37-m-dia antenna with 10 W of transmit power. Equally critical in implementing both links is mounting the antennas on a relay-bus-mounted "yoke" which requires twice per lunar cycle 180 degree repositioning to maintain the links, as shown in Figures 6a and 6b.

2. Sail Propulsion System

The Sail Propulsion System has a mass of 60 kg including 20% contingency. It uses inflatable sub-Tg space-rigidized boom technology[3] developed by L'Garde, Inc., Tustin, CA under the NASA ISPT

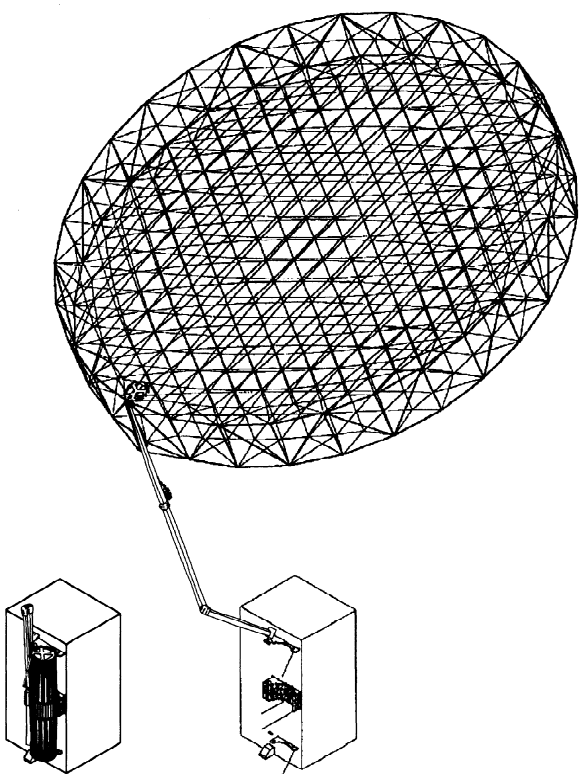


Figure 8. S-band Northrop Grumman AstroMesh antenna.



Figure 9. Deployed 20-m square sail propulsion system ground demonstrator in NASA Plum Brook vacuum chamber, 2004.

Program's FY 2003-2005 Solar Sail Ground Demonstrator Program and proposed for NASA New Millennium Program (NMP) Space Technology (ST)-9 flight demonstration by a proposal team led by the NASA Goddard Space Flight Center (GSFC) and comprised of team members from the Jet Propulsion Laboratory, L'Garde, Inc., and Orbital Sciences. The sail configuration for the Polesitter relay satellite is square, 165 meters on a side. The sail uses 0.9-micron-thick Mylar for the sail membrane. So sized, the SPS provides a maximum characteristic acceleration (a_0) of 1.2 mm/s2 to the 195-kg relay satellite to enable maintenance of the near-L2 operational station location. At 165 meters on-a-side, the sail size is similar to that required for the first NASA Science Roadmap mission requiring solar sail propulsion, the Solar Polar Imager. Required for the Polesitter mission application are two SPS features developed under the ISPT Program not proposed for the NMP ST-9 sail flight demonstration due to cost constraints: 1) a spreader system that adds rigidity to the sail to permit its scaling up from the 35-m on-a-side design proposed for the ST-9 flight demonstration to the 165-m on-a-side size required by the Polesitter, and 2) control vanes to permit autonomous, propellantless control of the relay satellite's attitude and position. Figure 9 shows the 20-m × 20-m square ground demonstrator version of the SPS in its fully deployed state in the NASA Plum Brook vacuum test chamber upon successful completion of the ISPT Program's Sail Ground Demonstrator Program in CY 2004 (Ref. 3).

3. Technology Readiness and Heritage

Figure 10 summarizes the key points bearing on the technology readiness of the elements comprising the Polesitter flight vehicle: the relay bus, carrier vehicle, and Sail Propulsion System. The metric used to assess readiness is the NASA 9-level Technology Readiness Level (TRL) scale, where "9" indicates the highest level of maturity, equivalent to an actual flight system having been proven through successful mission operations, and "1" indicates the lowest level of maturity, equivalent to basic principles having been observed and reported. As shown, and for the reasons noted, all elements of the relay bus and carrier vehicle are TRL 9, and the Sail Propulsion System is TRL 5-6. The relatively low TRL level for the SPS, with level 6 defined as a "system/subsystem model or prototype demonstration in a relevant environment (ground or space)," suggests the desirability of a risk reduction program to increase the TRL of the SPS, which this author supports, as acknowledged in paragraph **VI**.

- **Relay Bus and Carrier Vehicle**
  - All elements TRL 9.
  - Utilizes conventional high-flight-heritage small-sat satellite elements closely following the designs developed for the JPL-led New Millennium Program (NMP) Space Technology (ST)-5 and GSFC-led ST-9 flight demonstration proposals and the NASA ISPT Program's FY2003–2005 Solar Sail Ground Demonstrator Program.
  - Utilizes high-flight-heritage Northrop Grumman S-band AstroMesh antenna.

- **Sail Propulsion System (SPS)**
  - TRL 5-6.
  - Utilizes the inflatable sub-Tg space-rigidized boom technology developed by L'Garde, Inc., Tustin, CA in FY 2003-2005 under the ISPT Program's Sail Ground Demonstrator Program and proposed for NMP ST-9 flight demonstration by the GSFC-led GSFC/JPL/L'Garde, Inc./Orbital Sciences Corporation ST-9 proposal team.
  - Total ISPT Program Solar Sail Ground Demonstrator Program investment in solar sail technology: ~ $37.5 M ($17.5 M for the inflatable-boom technology specifically underpinning the Polesitter concept; $2.5 M for navigation tools, including, in specific the JPL-developed Solar Sail Spaceflight Simulation Software [S5] tool).
  - Risk reduction program appropriate and recommended to retire outstanding risks.

Figure 10. Polesitter technology readiness and heritage.

V. Opportunities for International Cooperation

World interest in solar sail technology is high and opportunities for international cooperation in the development of the Lunar Polesitter are many, as noted below. This circumstance merges well with NASA's interest in promoting interest in international cooperation in the Human Exploration Initiative (HEI) and specific use of an "open architecture" plan to encourage participation. Under the plan, the United States and NASA will build the transportation infrastructure, the initial surface mobility capabilities, and the initial basic mission support communications and navigation capabilities in support of HEI with international cooperation encouraged once the initial capabilities are established. Specifically noteworthy with regard to NASA's interest in cooperation in the communications area in general, and potentially a Lunar Polesitter communications satellite in specific, is its identification in the plan of high-bandwidth communications and navigation as areas for potential cooperation.

With regard to world interest in solar sail technology, and potential world interest in participation in a Lunar Polesitter development program, Russia, Japan, Germany, the United Kingdom, and Canada serve as the best examples. To those in the solar sail community, Russia's recent partnership with The Planetary Society in attempting to launch, albeit unsuccessfully, the first solar sail is well known. Also well known are Japan's recent first successful space deployment of a solar sail, Germany's longstanding support for solar sail technology development, particularly in booms and structures, and the United Kingdom's longstanding leadership role in solar sail mission and navigation studies. Less well known, but equally important, is Canada's potential interest in Earth polesitter and communications applications of solar sail technology and sail technology demonstration missions.

Areas this author sees as potentially fruitful for international cooperation in the development of the Lunar Polesitter relay satellite for which the United States could trade communications bandwidth from the Lunar surface for contributions-in-kind include:

1) Launch Vehicle: Potential donation from Russia, Japan, France, or the European Space Agency.
2) Relay Satellite: Bus components from France and Germany (star trackers/reaction wheels). Communications subsystem/equipment from Canada.
3) Sail Propulsion System: Sail membrane from Japan.
4) Mission Design/Navigation: Analysis and support from the United Kingdom.

VI. Notional Development Timeline

Figure 11 shows a notional development timeline for the Lunar Polesitter leading to launch of the first satellite in CY 2020. Key aspects of the timeline include detailed design studies of the relay satellite in the CY 2008–2010

Date (FY)	Activity/Decision
• **Now**–2008 end	• Concept refinement studies: - Mission, navigation, communications, SPS, and flight vehicle system design; coverage analysis; launch vehicle definition; cost. - Study portability to Mars.
• 2009	• Continue concept refinement studies. Explore partnering opportunities.
• 2010–2013	• Continue refinement studies. Develop partnerships. Define risk reduction program.
• 2013	NASA decision to implement risk reduction program.
• 2013–2016	• Implement risk reduction program: - Sail Propulsion System: Hardware and software risk reduction (sail-boom inflation system); sail deployment/performance prediction tool development and validation. - Navigation Tools: Trajectory design and analysis tool development and validation.
• 2016–2020	• NASA project start decision / project implementation.
* 2020	• Launch first relay satellite.
• 2025–2027	• Add additional, optional, enhanced-capability relay satellite (Complete lunar far-side coverage; higher-bandwidth: optical communications/other).
• 2030–2035?	• Port relay satellite to Mars (if viable).

Figure 11. Notional Polesitter development schedule.

timeframe, partnership development in the CY 2010–2013 timeframe, and implementation of a risk reduction program in the CY 2013–2016 timeframe. With requisite funding, launch in the CY 2017–2018 timeframe could be achieved without additional risk.

VII. Cost

The estimates completed in this study suggest design and development costs for the first Polesitter vehicle on the order of $275–300 million in fiscal year 2007 dollars for a four-year development program, including 30% reserves and excluding launch vehicle, launch and mission operations, and SPS risk reduction program costs.

VIII. Conclusion

The preliminary JPL/industry studies reported herein suggest the Lunar Polesitter Relay Satellite concept is viable, meets NASA's needs for communications and navigation at the Moon, and offers better coverage, lower resource demand on surface assets, greater operational simplicity, and lower cost than other alternatives. Should subsequent concept refinement studies confirm these initial promising results, a wholly new area for solar sail applications may emerge with the potential to contribute significantly to the nation's future goals in space exploration.

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